

Swept Wing Structural Design Study

Date Started: September 2026

Software: SOLIDWORKS 2026, SOLIDWORKS Simulation

Project Objective

This project started as a guided SOLIDWORKS modeling tutorial on YouTube for a swept-back wing. My original goal was to improve my SOLIDWORKS skills and become familiar with more modeling techniques. After completing the tutorial model, I used ChatGPT to help me learn SOLIDWORKS simulation and I used the wing as a baseline for an independent structural study to investigate how changes to the wing structure could affect its mass and stiffness.

Link to tutorial: <https://youtu.be/hb3aDPU26qE?si=gAoIAbooGgCOUHWf>

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What I Learned From Modeling

- Surface modeling
- Swept geometry
- Wing ribs
- Main and Rear Spars
- 3D Sketching
- Cutting with surfaces
- Extending/trimming surfaces

Baseline Model

Property	Baseline
Airfoil	NACA 4412
Semi-Span	~4436 mm
Root chord	~2019 mm
Tip chord	~808 mm
Material assumption	6061 aluminum
Mass	33.77 kg

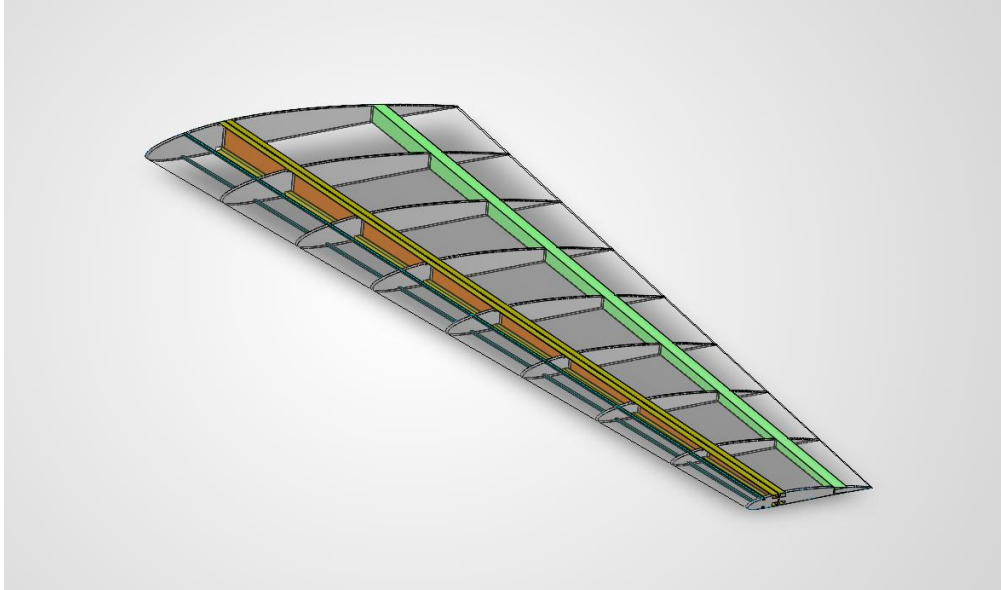


Figure 1. Completed swept-wing structural model.

Baseline Structural Study

Study type: Linear static finite element analysis (FEA) using SOLIDWORKS Simulation

Assumptions:

- Wing root treated as completely fixed
- Solid bodies treated as bonded
- 6061 aluminum used throughout baseline structure
- Reference surface excluded from structural analysis
 - The lofted wing surface was excluded from the structural analysis because it was modeled as a zero-thickness reference surface rather than a physical wing skin. The FEA therefore represents only the internal structural members.
- **5000 N total upward test load**
- Load distributed across five spanwise structural faces
- Load is a standardized comparison load, not a calculated aerodynamic flight load
- Default SOLIDWORKS mesh used

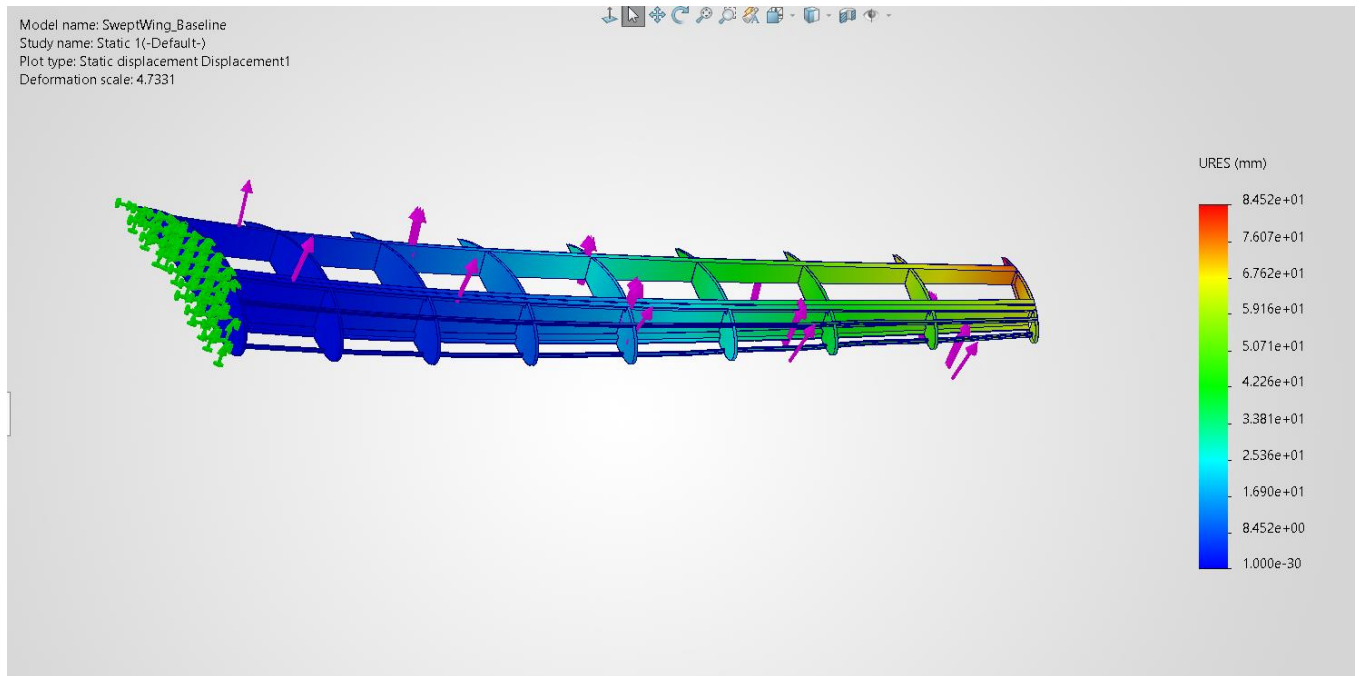


Figure 2. Baseline displacement under 5,000 N test load.

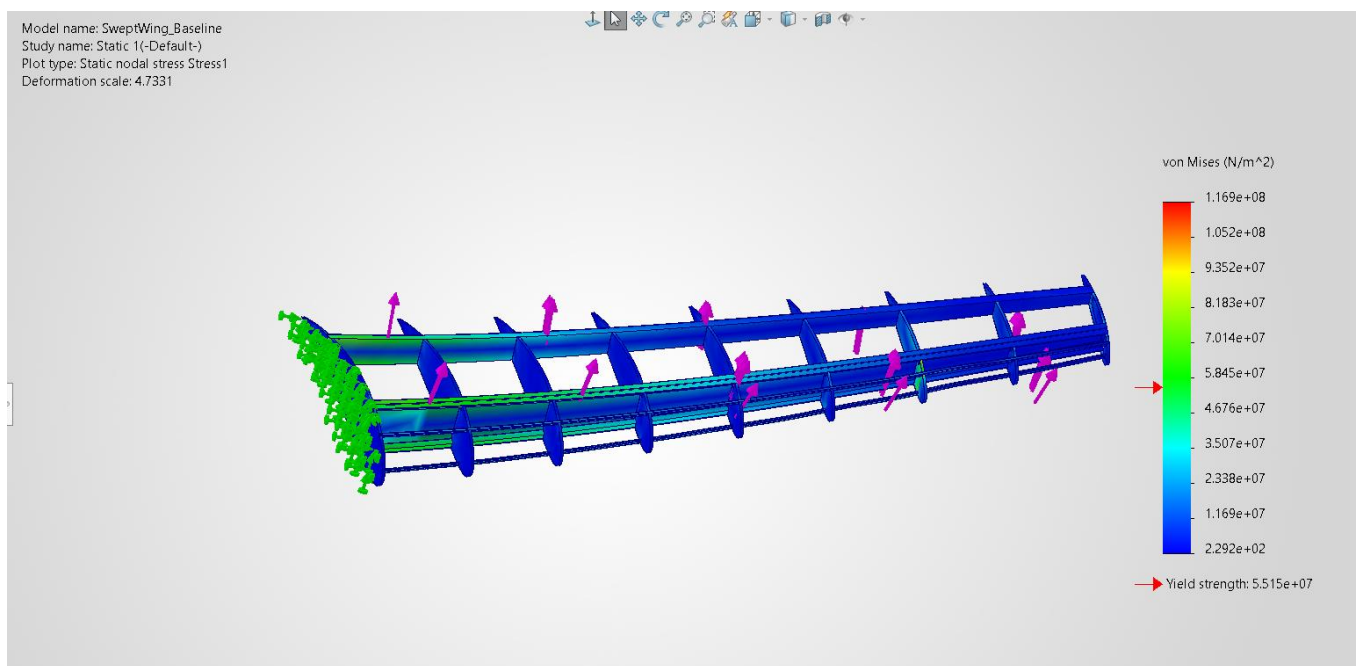


Figure 3. Baseline von Mises stress distribution under the 5,000 N test load.

Results:

Maximum resultant displacement	84.52 mm
Maximum von Mises stress	116.9 MPa

Independent Material Study

Research Question: Can a higher specific stiffness spar material reduce wing deflection while also reducing structural mass?

My hypothesis: Replacing the primary aluminum spars with a CFRP-like material will decrease the total wing mass and decrease maximum wing-tip displacement under the same loading conditions.

Material Research

This is what I gathered from researching about the baseline material and the material I want to use to study.

6061 aluminum

- Isotropic for our simplified analysis
- Has a defined yield behavior
- $\sim 2700 \text{ kg/m}^3$ density
- $\sim 69 \text{ GPa}$ elastic modulus

Carbon Fiber Reinforced Plastic (CFRP)

- Lower density
- Potentially very high longitudinal stiffness
- Direction-dependent properties
- Does not exhibit conventional ductile yielding like aluminum
- Failure can involve fiber fracture, etc.

Disclaimer: I did not model a real composite laminate.

Simplified Material:

Property	Assumed value
Density	1500 kg/m^3
Elastic Modulus	150 GPa
Poisson's ratio	0.30
Shear Modulus	57.7 GPa
Model	Linear elastic isotropic

Note: This material model was intentionally simplified for an introductory stiffness-to-weight comparison and should not be interpreted as a validated CFRP laminate model

Results

Metric	Aluminum Baseline	CFRP-like Spars	Change
Total wing mass	33.77 kg	27.04 kg	~19.9%
Maximum displacement	84.52 mm	60.78 mm	~28.1%
Applied test load	5000 N	5000 N	--

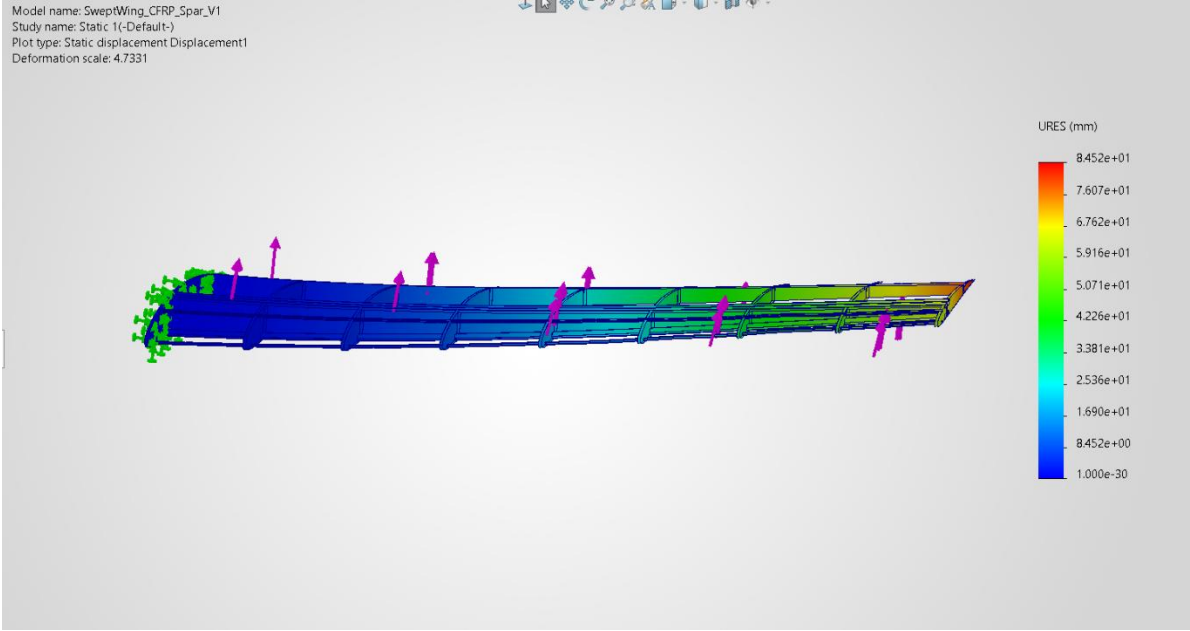


Figure 3. Baseline displacement under 5,000 N test load.

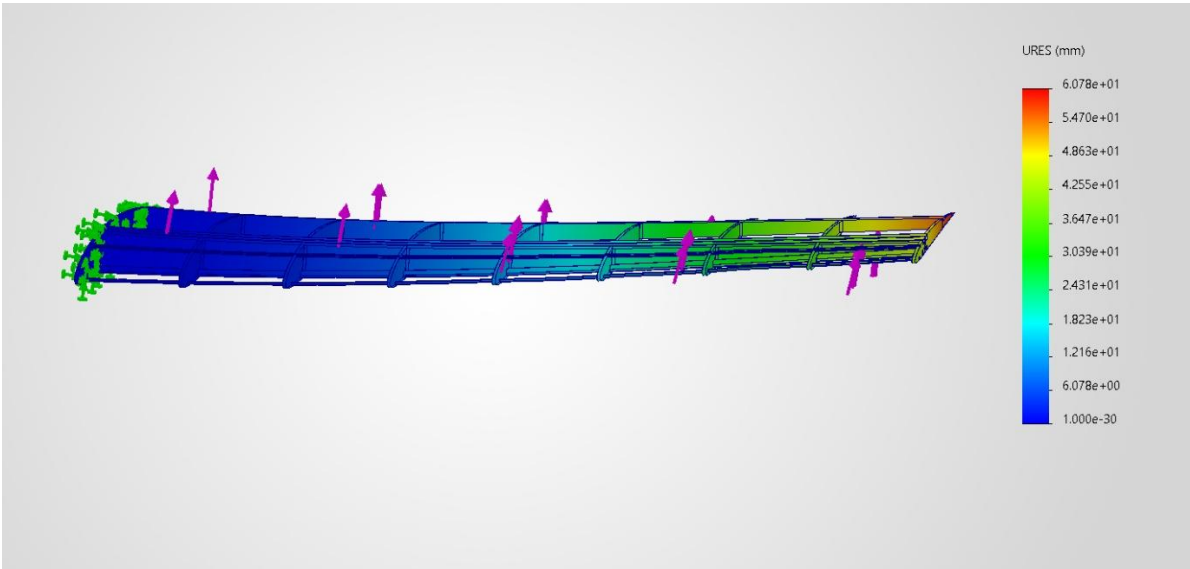


Figure 4. CFRP-like spar displacement under identical loading.

Conclusion

The results supported my original hypothesis. Under the assumptions of the simplified model, replacing the main and rear aluminum spars with the CFRP-like material reduced total structural mass by approximately 20% while reducing maximum displacement by approximately 28%.

These results represent a comparative study rather than a prediction of real aircraft performance. The loading was not derived from aerodynamic conditions, CFRP was approximated as isotropic, connections were idealized as bonded, the wing root was perfectly fixed, and the mesh convergence was not evaluated.

Future Work

- Model a wing skin with realistic thickness and include its structural contribution in the FEA.
- Perform a mesh-convergence study to evaluate result sensitivity to mesh size
- Replace the isotropic CFRP approximation with an orthotropic composite model.
- Investigate alternate spar geometries and material configurations.
- Apply a more realistic aerodynamic load distribution.
- Compare structural efficiency using mass, stiffness, and stress.